

Notes: Sufi (JF, 2007)

Information Asymmetry and Financing Arrangements: Evidence from Syndicated Loans

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1 Big picture

- **Question.** How does information asymmetry between borrowers and lenders affect the structure of syndicated loan contracts, especially the allocation of loan shares across lead arrangers and participant lenders?
- **Setting.** The paper studies the U.S. syndicated loan market from 1992 to 2003 using 12,672 loan deals made to 4,687 nonfinancial firms.
- **Key institutional feature.** A lead arranger negotiates the loan, performs due diligence, monitors the borrower, and then sells pieces of the loan to participant lenders. This creates a delegated monitoring problem: participants rely on the lead bank, but the lead bank may shirk if it retains too small a stake.
- **Main measure of borrower opacity.** Borrowers are classified as *private*, *unrated*, or *transparent*. Private firms lack public SEC filings; unrated public firms have SEC filings but no S&P senior unsecured debt rating; transparent firms have both SEC filings and a rating.
- **Main result.** When borrowers are more opaque, lead arrangers retain a larger share of the loan and form more concentrated syndicates with fewer participant lenders.
- **Mechanism.** Opaque borrowers require more due diligence and monitoring. A larger retained stake gives the lead arranger stronger incentives to exert effort and credibly assures participant lenders.
- **Borrower reputation result.** The opacity effect weakens when opaque borrowers repeatedly access the syndicated loan market. Relationship history partially substitutes for public information.
- **Lead-bank reputation result.** Reputable lead arrangers reduce information problems, but they do not fully eliminate the need for the lead to retain a larger stake.
- **Participant choice result.** When borrowers are opaque, participants are more likely to have prior lending relationships with the borrower and to be geographically closer to the borrower or lead arranger.
- **Overall takeaway.** Syndicate structure is not just a mechanical way to spread risk. It is an endogenous contracting response to adverse selection, monitoring incentives, borrower opacity, reputation, and relationship networks.

2 Incremental contribution of the paper

2.1 Contribution relative to banking and delegated monitoring theories

- Classic delegated monitoring theory argues that banks are valuable because they screen and monitor opaque borrowers.
- This paper goes inside a lending syndicate and studies a second layer of delegation: participant lenders delegate screening and monitoring to the lead arranger.
- The incremental contribution is to show that the lead arranger's retained stake is a measurable incentive device that mitigates moral hazard in monitoring.
- Instead of treating bank lending as a single lender-borrower relationship, the paper studies how multiple lenders organize themselves when monitoring effort is costly and noncontractible.

Interpretation: The paper therefore extends bank monitoring theory from the question “Why do banks lend?” to the question “How do banks share loans when one bank must monitor for many?”

2.2 Contribution relative to empirical debt contracting research

- Prior empirical work often studies pricing, covenants, maturity, collateral, or the choice between public and private debt.
- Sufi studies the internal organization of the loan syndicate: number of lenders, lead share, concentration, and participant identity.
- This provides direct evidence on contracting responses to information asymmetry that is not visible from loan spreads alone.
- The paper shows that opacity changes not only loan terms but also the allocation of monitoring incentives across lenders.

Interpretation: The contribution is important because two loans with similar spreads can have very different syndicate structures and therefore very different incentive arrangements.

2.3 Contribution relative to relationship lending and reputation literatures

- The paper shows that repeated borrowing reduces the need for lead arranger retention among opaque firms.
- It also shows that lead arranger reputation mitigates but does not eliminate information problems.
- Participant banks are more likely to join opaque deals when they have previous borrower relationships or are located close to the borrower.
- These findings connect relationship lending and reputation to the market design of syndicated loans.

Interpretation: The key contribution is not just that relationships matter; it is that relationships shape which lenders are chosen and how much exposure the lead arranger must keep.

2.4 Contribution relative to the literature on financial intermediation markets

- Syndicated lending is a large source of corporate finance, but it had received relatively little empirical attention compared with public bonds and equity.
- The paper provides one of the first broad empirical analyses of syndicated loan syndicate structure using deal-level data.
- It documents systematic variation in the organization of syndicates across borrower opacity, borrower reputation, lead-bank reputation, collateral, and participant characteristics.

Interpretation: The incremental contribution is to make syndicated loans an empirical laboratory for testing theories of information asymmetry, monitoring, and financial intermediation.

3 Data and measurement

3.1 Syndicated loan data

- The sample covers 12,672 syndicated loan deals from 1992 through 2003.
- The deals are made to 4,687 borrowing firms.
- The data come from syndicated loan sources that identify lead arrangers, participant lenders, loan amounts, loan purposes, maturities, collateral, term loans, tranches, and syndicate structure.

- The sample includes a broad range of credit quality: private firms, unrated public firms, and rated public firms.

Interpretation: The syndicated loan market is ideal for studying information asymmetry because both very opaque and very transparent borrowers obtain syndicated loans.

3.2 Borrower opacity categories

Borrowers are divided into three categories:

- **Private firms:** firms with no publicly available SEC filings.
- **Unrated firms:** firms with publicly available SEC filings but no S&P senior unsecured debt rating.
- **Transparent firms:** firms with both public SEC filings and an S&P senior unsecured debt rating.

Interpretation: The ordering captures increasing transparency. Private firms are hardest to evaluate and monitor; transparent firms have both regulatory disclosure and third-party credit evaluation.

3.3 Syndicate structure outcomes

The main outcomes are:

- total number of lenders;
- total number of lead arrangers;
- total number of participant lenders;
- percentage of the loan retained by each lead arranger;
- dollar amount retained by each lead arranger;
- exposure of the lead arranger measured in basis points;
- Herfindahl concentration index of lender shares.

Interpretation: These outcomes measure how much monitoring exposure the lead arranger keeps and how dispersed the syndicate is across participant lenders.

3.4 Borrower and loan controls

The paper controls for:

- firm size, measured by sales;
- loan amount, including flexible loan-size splines;
- loan maturity;
- whether the loan has more than one tranche;
- whether the deal includes a term loan;
- loan purpose: working capital/corporate purposes, refinancing, acquisition, backup line, or other;
- year and industry fixed effects.

Interpretation: These controls are crucial because private and unrated borrowers tend to be smaller and obtain smaller loans. The paper’s estimates should therefore be read as conditional comparisons among similar loans and firms.

3.5 Participant lender data

For participant-choice tests, the paper forms potential participant-bank observations and measures whether the lender is chosen for a deal. Explanatory variables include:

- whether the participant previously led a loan to the borrower;
- whether the participant previously participated in a loan to the borrower;
- whether the participant is in the same region as the borrower;
- whether the participant is in the same region as the lead arranger;
- bank size and capital measures.

Interpretation: These variables test whether relationship and geographic proximity substitute for public borrower information.

4 Conceptual framework

4.1 Lead arranger moral hazard

The lead arranger performs costly screening and monitoring, but participant lenders cannot perfectly observe this effort. If the lead arranger sells too much of the loan, it bears too little downside risk and may underinvest in due diligence and monitoring.

$$\text{Monitoring Incentive} \uparrow \quad \text{when} \quad \text{Lead Share} \uparrow. \quad (1)$$

Interpretation: Retaining a larger loan share is a bonding mechanism. It convinces participants that the lead arranger has enough skin in the game.

4.2 Prediction 1: borrower opacity changes syndicate structure

When the borrower is more opaque:

$$\text{Opacity} \uparrow \Rightarrow \text{Lead Share} \uparrow, \quad \text{Syndicate Concentration} \uparrow, \quad \text{Participants} \downarrow. \quad (2)$$

Interpretation: More opacity increases the value of monitoring and the severity of lead-arranger moral hazard, so the lead must keep more exposure.

4.3 Prediction 2: reputation and repeat borrowing reduce opacity problems

If the borrower repeatedly borrows in the syndicated loan market, lenders learn about the borrower. If the lead arranger has a strong reputation, participants trust the lead more.

$$\text{Borrower Reputation} \uparrow \Rightarrow \text{Lead Share Required} \downarrow. \quad (3)$$

$$\text{Lead Reputation} \uparrow \Rightarrow \text{Information Frictions} \downarrow. \quad (4)$$

Interpretation: Reputation reduces information asymmetry, but it need not eliminate the need for incentive alignment.

4.4 Prediction 3: participant identity is endogenous

When borrowers are opaque, participants with superior information are more valuable. Thus opaque deals should attract:

- lenders with prior relationships with the borrower;
- lenders geographically close to the borrower;
- lenders close to the lead arranger, if lead-participant communication matters.

Interpretation: Syndicate composition is part of the contracting solution. The lead arranger can reduce information frictions by choosing informed or proximate participants.

5 Empirical specifications

5.1 Syndicate structure regressions

The baseline specification is:

$$SyndicateStructure_i = \alpha + \gamma Opaque_i + X_i' \beta + Year_t + Industry_j + \varepsilon_i. \quad (5)$$

The dependent variable can be number of participants, lead share, Herfindahl concentration, or lead exposure. Transparent firms are the omitted group.

Interpretation: A positive coefficient on private or unrated status in lead-share or concentration regressions means that opacity changes syndicate structure in the direction predicted by monitoring theory.

5.2 Repeat borrower regressions

The paper estimates:

$$Y_i = \alpha + \beta_1 Opaque_i + \beta_2 \ln(1 + PreviousLoans_i) + \beta_3 Opaque_i \times \ln(1 + PreviousLoans_i) + X_i' \delta + \varepsilon_i. \quad (6)$$

Interpretation: A negative interaction coefficient means that repeat borrowing attenuates the effect of opacity because the borrower becomes known to the market.

5.3 Lead reputation regressions

Lead reputation is measured by lead arranger market share in the prior year. The paper estimates whether opacity effects are weaker when a top lead arranger runs the syndicate.

Interpretation: If reputation substitutes for retained stake, the interaction between opacity and lead reputation should reduce lead retention and concentration.

5.4 Alternative explanation tests

The paper tests whether the results reflect:

- risk sharing rather than monitoring incentives;
- adverse selection in loan sales rather than moral hazard in monitoring;
- collateral or secured status;
- firm size rather than opacity;
- public firms' need for monitoring, measured by accruals, R&D, leverage, and related proxies.

Interpretation: These tests are meant to show that the core effect is not merely a loan-size, credit-risk, or diversification result.

5.5 Participant choice probit regressions

For potential participants, the model is:

$$Pr(Chosen_{b,i} = 1) = \Phi(\alpha + \beta_1 Relationship_{b,i} + \beta_2 Geography_{b,i} + \beta_3 Opaque_i + X'_{b,i}\gamma). \quad (7)$$

Interpretation: The key prediction is that relationship and geography matter more when borrowers are private or unrated.

6 Identification and empirical design

6.1 Main identifying variation

- The paper compares syndicate structure across borrowers with different levels of transparency.
- It conditions on loan size, firm size, maturity, tranching, term-loan status, purpose, year, and industry.
- The design asks whether private and unrated borrowers have different syndicate structures than rated public borrowers, even after these controls.

Interpretation: The identifying assumption is that, conditional on controls, private/unrated status captures information asymmetry rather than unrelated differences in loans.

6.2 Why multiple tests matter

- The syndicate-structure regressions show that opaque borrowers lead to more lead retention and concentrated syndicates.
- The repeat-borrower tests show that information accumulates over time.
- The lead-reputation tests show that intermediary reputation matters.
- The secured/unsecured and public-firm monitoring tests separate opacity from collateral and firm size.
- The participant-choice tests show that participants are selected based on information-relevant relationships and proximity.

Interpretation: No single test proves the mechanism. The paper’s strength is that many empirical patterns line up with the same information-asymmetry theory.

6.3 Limitations

- Borrowers are not randomly assigned to be private, unrated, or rated.
- Private and unrated status may correlate with unobserved borrower risk.
- Syndicate structure is jointly determined with loan terms such as pricing, collateral, maturity, and covenants.
- The paper does not directly observe monitoring effort.
- Nevertheless, the retained-stake and participant-selection patterns are difficult to explain without information frictions.

Interpretation: The evidence is strongest as a rich set of contracting patterns consistent with monitoring incentives, rather than as a single randomized causal estimate.

7 Tables and figures

7.1 Tables I and II: sample and syndicated loan market structure

Read:

- Table I reports 12,672 deals and 4,687 firms. The average deal size is \$364 million, and the median is \$150 million.
- On average, syndicates have 8.1 lenders, 1.7 lead arrangers, and 6.4 participant lenders.
- Lead arrangers retain 28.5% of the loan on average in the subsample with lender shares.
- Table II shows that the syndicated loan market is dominated by large lead arrangers, but participant composition varies across borrower types.

Economic message: The market is large, organized around repeat lead arrangers, and has enough variation in borrower opacity and syndicate design to test information-asymmetry theories.

Table I

Summary Statistics for Syndicated Loan Deals

This table presents summary statistics for the sample of 12,672 syndicated loan deals representing 4,687 firms from 1992 through 2003. Summary statistics of firm characteristics are calculated at the firm level, and summary statistics for deal characteristics are calculated at the deal level.

	No. of Firms	Mean	SD	Distribution		
				10 th	50 th	90 th
Firm characteristics						
Total sales (\$M)	4,687	1,797	6,925	48	367	3,490
Number of previous syndicated loans	4,687	1.12	1	0	1	3
S&P debt rating indicator variable	4,687	0.31	0.43			
Compustat data available						
Total assets (book value, \$M)	2,968	3,320	12,174	105	600	7,063
Leverage (book debt/book assets)	2,968	0.34	0.20	0.10	0.32	0.6
Earnings to assets	2,968	0.07	0.07	0.00	0.08	0.15
Lead arranger characteristics						
Total assets (book value, \$B) ^a	12,672	333	232	70	262	694
Market share, previous year ^a	12,672	0.09	0.10	0.00	0.05	0.27
Syndicated loan characteristics						
Size of deal (\$M)	12,672	364	696	40	150	8,50
Maturity (days)	12,672	1,103	755	364	1,095	2,008
> 1 loan tranche indicator variable	12,672	0.26	0.44			
Deal includes term loan indicator	12,672	0.22	0.42			
Spread on drawn funds	10,898	159	123	31	125	325
Syndicate structure						
Total number of lenders	12,672	8.1	7.7	2	5	18
Total number of lead arrangers	12,672	1.7	2.1	1	1	3
Total number of participant lenders	12,672	6.4	7.3	1	4	16
% kept by lead arranger ^a	4,414	28.5	19.3	8.0	23.5	55.6
Amount kept by lead arranger (\$M) ^a	4,414	48.4	70.4	13.0	31.9	91.3
Exposure of lead (basis points) ^a	4,414	0.03	0.04	0.00	0.02	0.08
Concentration of syndicate (Herfindahl)	4,414	2,383	1,759	590	1,844	5,001
Purpose of loan indicator						
Working capital/corporate purposes	12,672	0.42	0.49			
Refinancing	12,672	0.27	0.44			
Acquisitions	12,672	0.14	0.35			
Backup line	12,672	0.11	0.31			
Other	12,672	0.07	0.25			

^aRepresents average of lead arrangers when there is more than one lead on the loan (31% of

Table I: Summary statistics

Table II
Top Lead Arrangers and Participant Banks, by Market

This table lists the top five lead arrangers (by deal amount) and top five participants (by total number of deals) for syndicated loans in the sample from 2001 to 2003. Market share figures for lead arrangers split the amount of a given syndicated loan equally over all lead arrangers when there are multiple leads. Firms are considered "private" if they have no publicly available SEC filings. The third column reports the most common participant or the top lead arrangers (and the fraction of all deals in which they participate with the lead arranger) on deals with exactly one lead arranger.

(1) Loans to Private Firms		(2) Loans to Public Firms		(3) Most Common Participant	
Lead arrangers	Mkt. share		Mkt. share		% Lead deals
Bank of America	0.23	JPMorganChase	0.34	Bank of America	
JPMorganChase	0.18	Citigroup	0.19	Fleet	20
Citigroup	0.10	Bank of America	0.16		
Bank One	0.05	Bank One	0.08	JPMorganChase	
Fleet	0.05	Deutsche Bank	0.02	Bank of America	36
Total amount (\$B)	157	Total amount (\$B)	1,373	Citigroup	
Market Herfindahl	1,077	Market Herfindahl	1,872	JPMorganChase	49
Participants				Bank One	
U.S. Bancorp	# deals		# deals	U.S. Bancorp	27
ARN-AMRO	148	Fleet	772		
Fleet	134	Bank of Tokyo-Mitsubishi	769	Fleet	
Wachovia	133	Bank One	707	Wachovia	24
National City	118	Wachovia	707		
Bank One	115	Bank of America	706	Deutsche Bank	
				Fleet	35
Total number of participants	3,448	Total number of participants	23,922		

Table II: Top lead arrangers and participants

7.2 Tables III and IV: opacity and syndicate structure

Read:

- Table III shows that private and unrated firms have fewer lenders, fewer participant banks, higher lead shares, and higher syndicate concentration than transparent firms.
- Lead arrangers hold 38.3% of private-firm loans, 35.3% of unrated-firm loans, and 19.6% of transparent-firm loans.
- Table IV confirms the pattern in regressions with controls for firm size, loan size, maturity, tranches, term loans, purpose, year, and industry.
- Private borrowers have fewer participants and higher lead exposure; unrated borrowers show similar but weaker effects.

Economic message: Borrower opacity changes syndicate structure in exactly the way predicted by moral hazard in monitoring: the lead keeps more skin in the game.

Cell Means, by Measure of Information Asymmetry

This table presents cell means and standard errors, by group, for the sample of 12,672 syndicated loan deals representing 4,687 firms from 1992 through 2003. A "private" firm is a firm with no publicly available SEC filings. An "unrated" firm is a firm with publicly available SEC filings and no S&P senior unsecured debt rating. A "transparent" firm is a firm with publicly available SEC filings and an S&P senior unsecured debt rating.

	Private	Unrated	Transparent
Percentage of sample loans	0.21	0.32	0.47
Total sales (\$M)	822 (74)	1090 (81)	5858 (171)
Size of deal (\$M)	156 (7)	174 (5)	590 (12)
Maturity (days)	1,070 (16)	1,177 (11)	1,067 (10)
> 1 tranche indicator variable	0.30 (0.01)	0.26 (0.01)	0.25 (0.01)
Deal includes term loan	0.31 (0.01)	0.23 (0.01)	0.18 (0.01)
Syndicate structure characteristics			
Total number of lenders	4.77 (0.09)	6.09 (0.08)	11.05 (0.12)
Total number of lead arrangers	1.42 (0.03)	1.51 (0.02)	2.05 (0.03)
Total number of participant banks	3.35 (0.08)	4.58 (0.08)	9.01 (0.11)
% of loan kept by each lead (avg)	38.3 (0.8)	35.3 (0.4)	19.6 (0.4)
Smallest 1/3 firms, by sales	43.5 (1.1)	41.8 (0.6)	29.3 (1.1)
Middle 1/3 firms, by sales	35.2 (1.4)	31.8 (0.6)	23.5 (0.6)
Largest 1/3 firms, by sales	21.8 (2.0)	20.2 (1.0)	15.2 (0.4)
Conc. of syndicate (Herfindahl)	3,274 (75)	2,990 (40)	1,580 (32)
Smallest 1/3 firms, by sales	3,742 (96)	3,587 (54)	2,415 (102)
Middle 1/3 firms, by sales	2,991 (124)	2,640 (59)	1,928 (60)
Largest 1/3 firms, by sales	1,763 (163)	1,729 (95)	1,199 (36)

**Table IV
Information Asymmetry Syndicate Structure Regressions**

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. A "private" firm is a firm with no publicly available SEC filings, an "unrated" firm is a firm with publicly available SEC filings and no S&P senior unsecured credit rating, and a "transparent" firm is a firm with publicly available SEC filings and a credit rating. To control for loan size, the sample is split into third based on size, and the interest (not reported) and natural log of loan size (reported) varies for each group. All regressions include year and industry dummies, and standard errors are heteroskedasticity robust, clustered at the borrowing firm. "Transparent" firms are the omitted group. "Corporate purposes" is the omitted loan purpose group.

Dependent Variable	Full Sample		Loans for Which the Amount Held by Each Syndicate Member Is Available					
	(1) # Leads	(2) # Parts	(3) # Leads	(4) # Parts	(5) Herfindahl	(6) % Held By lead	(7) Amount Held By lead (\$M)	(8) Lead Exposure
Private firm	0.09 (0.09)	-1.62** (0.17)	-0.09 (0.08)	-0.71* (0.29)	259** (84)	2.94** (0.93)	6.30** (2.26)	0.0093* (0.0028)
Unrated firm	-0.02 (0.05)	-0.82** (0.15)	-0.04 (0.06)	-0.61** (0.20)	211** (57)	2.14** (0.62)	4.32* (1.77)	0.0078** (0.0017)
Ln(firm sales)	-0.01 (0.02)	0.15** (0.05)	0.01 (0.03)	0.56** (0.10)	-102** (23)	-1.32** (0.25)	-3.45** (0.97)	-0.0027** (0.0007)
Ln(loan amount)	0.22** (0.04)	0.91** (0.16)	0.23** (0.05)	0.78** (0.31)	-868** (83)	-7.69** (0.90)	11.14** (1.07)	0.0110** (0.0014)
Ln(loan amount) × middle	0.48** (0.09)	1.49** (0.25)	0.52** (0.09)	2.73** (0.37)	-144 (109)	-2.51* (1.22)	4.46* (1.94)	-0.0031 (0.0028)
Ln(loan amount) × large	0.15 (0.15)	2.86** (0.34)	0.78** (0.17)	4.47** (0.45)	665** (81)	6.19** (0.89)	72.66** (9.33)	0.0104** (0.0025)
Ln(maturity, in days)	0.00 (0.01)	0.43** (0.03)	0.07 (0.03)	0.36** (0.08)	-131** (24)	-1.12** (0.30)	-4.06** (1.01)	-0.0006 (0.0006)

(continued)

Table IV—Continued

Dependent Variable	Full Sample		Loans for Which the Amount Held by Each Syndicate Member Is Available					
	(1) # Leads	(2) # Parts	(3) # Leads	(4) # Parts	(5) Herfindahl	(6) % Held By lead	(7) Amount Held By lead (\$M)	(8) Lead Exposure
> 1 tranche indicator	0.03 (0.08)	0.33 (0.19)	0.00 (0.09)	-0.05 (0.24)	74 (60)	0.79 (0.62)	-5.18 (2.79)	0.0024 (0.0018)
Term tranche indicator	-0.04 (0.06)	0.56** (0.21)	-0.04 (0.09)	0.73** (0.27)	211** (79)	1.43 (0.85)	2.95 (2.51)	-0.0030 (0.0019)
Purpose: refinancing	0.05 (0.06)	0.86** (0.14)	0.14 (0.07)	0.25 (0.18)	0 (53)	-0.27 (0.57)	0.86 (1.34)	-0.0021 (0.0019)
Purpose: acquisition	0.22** (0.08)	0.60** (0.22)	0.41** (0.13)	-0.28 (0.32)	68 (73)	0.19 (0.82)	7.94* (3.65)	0.0043 (0.0025)
Purpose: backup	-0.39** (0.07)	0.94** (0.25)	-0.63** (0.12)	1.24** (0.36)	-203** (62)	-2.05** (0.86)	-11.20** (4.46)	-0.0067** (0.0022)
Purpose: other	0.15* (0.06)	0.53* (0.22)	0.19 (0.16)	0.33 (0.39)	256* (131)	1.46 (1.39)	10.37 (7.34)	0.0008 (0.0029)
N	12,672	12,672	4,414	4,414	4,414	4,414	4,414	4,414
R ²	0.22	0.37	0.19	0.55	0.48	0.31	0.14	0.14

**Significant at the 1% level, *significant at the 5% level, °coefficients distinct from each other at the 5% level.

Table IV: syndicate-structure regressions

Table IV continued

7.3 Tables V and VI: borrower reputation and lead-bank reputation

Read:

- Table V shows that opaque borrowers require higher lead retention when they are new to the syndicated loan market.
- The interaction between opacity and previous loans is negative: repeated borrowing reduces the incremental

lead stake required by opacity.

- Table VI shows that lead arranger reputation, measured by prior market share, reduces information problems.
- Still, reputation does not fully eliminate the need for lead retention.

Economic message: Reputation substitutes partially for public information. Borrower and lead-bank histories lower the cost of syndication, but incentive alignment remains important.

Repeat Borrowers Reduce Problems of Information Asymmetry

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. "Opaque" firms include both private and unrated firms. Transparent firms are the omitted group. In addition to variables reported, all regressions include year, industry, loan amount, and loan purpose dummies. Standard errors are heteroskedasticity robust, clustered at the firm level.

Dependent Variable	% Held by Lead			Herfindahl		
	(1)	(2)	(3)	(4)	(5)	(6)
Opaque firm	2.30** (0.60)	2.01** (0.62)	3.55** (0.85)	220** (55)	202** (56)	324** (77)
Ln[1+ # previous loans by firm]		-1.46** (0.41)	-0.60 (0.48)		-93* (38)	-24 (46)
Opaque firm × Ln[1+ # previous loans by firm]			-1.85* (0.76)			-148* (73)
Ln[firm sales]	-1.34** (0.25)	-1.22** (0.25)	-1.21** (0.25)	-103** (23)	-95** (23)	-94** (23)
Ln[loan amount]	-7.69** (0.90)	-7.65** (0.90)	-7.51** (0.90)	-809** (83)	-806** (83)	-795** (83)
Ln[loan amount] × middle	-2.50* (1.22)	-2.42* (1.22)	-2.49* (1.21)	-143 (109)	-138 (109)	-143 (108)
Ln[loan amount] × large	6.21** (0.89)	6.24** (0.88)	5.95** (0.89)	667** (81)	668** (81)	645** (82)
Ln[maturity, in days]	-1.13** (0.30)	-1.20** (0.30)	-1.19** (0.30)	-132** (24)	-136** (24)	-135** (24)
> 1 tranche indicator	0.77 (0.62)	0.63 (0.62)	0.59 (0.62)	73 (60)	64 (60)	61 (60)
Term tranche indicator	1.47 (0.86)	1.66 (0.85)	1.68* (0.85)	213** (79)	225** (79)	227** (79)
N	4,414	4,414	4,414	4,414	4,414	4,414
R ²	0.51	0.52	0.52	0.51	0.51	0.52

**Significant at the 1% level, *significant at the 5% level.

variable implies that the lead arranger retains 3.55% more of the loan when the borrower is opaque and has never accessed the syndicated loan market

Table V: repeat borrowers

Lead Bank Reputation Reduces Problems of Information Asymmetry

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. An "opaque" firm is a private firm or a firm with publicly available SEC filings and no S&P senior unsecured credit rating. A "transparent" firm is a firm with publicly available SEC filings and a credit rating. In all specifications, "transparent" firms are the omitted group. In addition to variables reported, all regressions include year, industry, loan amount, and loan purpose dummies. Standard errors are heteroskedasticity robust, clustered at the firm level.

Dependent Variable	% Held by Lead		Herfindahl	
	(1)	(2)	(3)	(4)
Opaque firm	3.51** (0.85)	4.25** (0.92)	320** (77)	355** (85)
Market share of lead arranger in prior year	-10.33** (2.99)	-3.88 (3.67)	-1071** (274)	-766* (320)
Opaque × Market share of lead arranger		-13.36* (5.48)		-631 (493)
Ln[1+ # previous loans by firm]	-0.57 (0.48)	-0.69 (0.48)	-22 (45)	-27 (45)
Opaque × Ln[1+ # previous loans by firm]	-1.84* (0.76)	-1.58* (0.77)	-147* (73)	-134 (74)
Ln[firm sales]	-1.18** (0.25)	-1.16** (0.25)	-91** (23)	-90** (23)
Ln[loan amount]	-7.57** (0.90)	-7.54** (0.90)	-801** (83)	-799** (83)
Ln[loan amount] × middle	-2.36 (1.21)	-2.43* (1.21)	-130 (109)	-134 (109)
Ln[loan amount] × large	6.19** (0.90)	6.00** (0.91)	671** (82)	662** (83)
Ln[maturity, in days]	-1.19** (0.30)	-1.18** (0.30)	-135** (24)	-135** (24)
> 1 tranche indicator	0.59 (0.61)	0.54 (0.61)	62 (60)	59 (60)
Term tranche indicator	1.66* (0.84)	1.65 (0.84)	224** (79)	224** (79)
N	4,414	4,414	4,414	4,414
R ²	0.52	0.52	0.52	0.52

**Significant at the 1% level, *significant at the 5% level.

Table VI: lead-bank reputation

7.4 Tables VII and VIII: alternative mechanisms and collateral

Read:

- Table VII evaluates whether the results are better explained by adverse selection or risk-sharing motives.
- The evidence is more consistent with moral hazard in monitoring than with pure risk sharing.
- Table VIII splits loans into secured and unsecured samples.
- The opacity effect appears in both samples, suggesting collateral does not fully solve information problems.

Economic message: Collateral and risk sharing matter, but the basic syndicate-structure response to information asymmetry remains visible after accounting for these channels.

Moral Hazard versus Adverse Selection

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. An "opaque" firm is a private firm or a firm with publicly available SEC filings and no S&P senior unsecured credit rating. A "transparent" firm is a firm with publicly available SEC filings and a credit rating. In all specifications, "transparent" firms are the omitted group. In addition to the variables reported, all regressions include year, industry, loan amount, and loan purpose dummies. Standard errors are heteroskedasticity robust, clustered at the firm level.

Dependent Variable	% Held by Lead		Herfindahl	
	(1)	(2)	(3)	(4)
Opaque firm	3.98** (0.93)	3.18** (1.07)	326** (86)	281** (97)
Lead is former lead for borrower	-2.03** (0.71)	-3.40** (1.11)	-215** (65)	-293** (103)
Opaque × Lead is former lead for borrower		2.12 (1.43)	121 (131)	
Market share of lead arranger in prior year	-3.53 (3.65)	-3.32 (3.65)	-729* (319)	-717* (318)
Opaque × Market share of lead arranger	-13.07* (5.47)	-13.36* (5.47)	-600 (491)	-617 (492)
Ln[1+ # previous loans by firm]	-0.01 (0.53)	0.47 (0.62)	45 (51)	73 (61)
Opaque × Ln[1+ # previous loans by firm]	-1.29 (0.77)	-2.12* (1.01)	-103 (73)	-151 (96)
Ln[firm sales]	-1.18** (0.25)	-1.19** (0.25)	-92** (23)	-93** (23)
Ln[loan amount]	-7.52** (0.89)	-7.52** (0.89)	-797** (82)	-797** (82)
Ln[loan amount] × middle	-2.49* (1.21)	-2.48* (1.20)	-139 (108)	-139 (108)
Ln[loan amount] × large	5.89** (0.90)	5.84** (0.90)	650** (83)	647** (83)
Ln[maturity, in days]	-1.17** (0.30)	-1.17** (0.30)	-134** (24)	-134** (24)
> 1 tranche indicator	0.55 (0.61)	0.55 (0.61)	60 (60)	60 (60)
Term tranche indicator	1.57 (0.84)	1.54 (0.84)	216** (78)	214** (78)

Table VII: moral hazard versus adverse selection

Secured versus Unsecured Loans and Information Asymmetry

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. An "opaque" firm is a private or unrated firm, and "transparent" firms are the omitted group. Columns (1) and (3) examine secured loans, and columns (2) and (4) examine unsecured loans. In addition to variables reported, all regressions include year, industry, loan amount, and loan purpose dummies. Standard errors are heteroskedasticity robust, clustered at the firm level.

Dependent Variable	% Held by Lead		Herfindahl	
	(1) Secured Loans	(2) Unsecured Loans	(3) Secured Loans	(4) Unsecured Loans
Opaque	1.44 (1.03)	2.70** (0.78)	186* (93)	218** (66)
Ln[firm sales]	-1.52** (0.45)	-0.84* (0.36)	-108** (41)	-64* (32)
Ln[loan amount]	-5.47** (1.00)	-15.23** (2.19)	-605** (88)	-1588** (208)
Ln[loan amount] × middle	-3.06 (1.96)	4.45 (2.51)	-243 (176)	677** (235)
Ln[loan amount] × large	4.36** (1.44)	13.20** (2.16)	488** (128)	1429** (205)
Ln[maturity, in days]	-0.92 (0.58)	-0.62 (0.49)	-126** (52)	-122** (44)
> 1 tranche indicator	2.74* (1.25)	0.64 (0.73)	214* (125)	49 (68)
Term tranche indicator	-0.73 (1.33)	2.56 (1.78)	95 (130)	213 (152)
N	1,797	1,237	1,797	1,237
R ²	0.39	0.62	0.39	0.64

**Significant at the 1% level, *significant at the 5% level.

Table VIII: secured versus unsecured loans

7.5 Tables IX and X: monitoring needs and participant characteristics

Read:

- Table IX focuses on public firms and uses measures such as accruals, R&D, leverage, and interest coverage to capture monitoring needs.
- Public firms that require more monitoring show higher lead retention and more concentrated syndicates.
- Table X shows that participant characteristics differ systematically by borrower transparency.
- For opaque firms, participants are more likely to have prior relationships and be geographically closer to the borrower or lead arranger.

Economic message: The evidence supports the interpretation that information problems are not only about private status. Even among public firms, monitoring needs shape syndicate design.

Information Asymmetry versus Size of Firm

This table reports coefficient estimates from regressions relating syndicate structure to information asymmetry of the borrower. Columns (1) and (2) examine all private and unrated borrowers ("opaque") and only transparent borrowers with total sales less than the median. Columns (3) through (6) examine only public firms using alternative measures of opacity. In addition to variables reported, all regressions include year, industry, loan amount, and loan purpose dummies. Standard errors are heteroskedasticity robust, clustered at the borrowing firm.

Dependent Variable	All Opaque + Smallest 1/2 Transparent, By Total Sales		Public Firms with Accrual Data Available		Public Firms with R&D Data Available	
	(1) % Held by Lead	(2) Herfindahl	(3) % Held by Lead	(4) Herfindahl	(5) % Held by Lead	(6) Herfindahl
Private firm	3.00** (0.96)	274** (87)				
Unrated firm	2.24** (0.67)	226** (61)				
Accruals to assets			8.42* (3.39)	658* (299)		
R&D to assets					16.32 (9.83)	2315* (919)
Income to assets			-3.51 (4.52)	-648 (419)	-5.69 (5.71)	-687 (516)
Leverage ratio			-1.32 (1.54)	-17 (142)	2.76 (2.17)	334 (198)
Ln[firm sales]	-1.82** (0.30)	-152** (27)	-1.39** (0.29)	-106** (26)	-1.39** (0.40)	-105** (35)
Ln[loan amount]	-7.00** (0.91)	-750** (82)	-9.26** (1.01)	-982** (88)	-9.91** (1.47)	-1023** (133)
Ln[loan amount] × middle	-2.65* (1.29)	-183 (114)	0.08 (1.35)	94 (118)	0.51 (1.90)	207 (168)
Ln[loan amount] × large	5.49** (1.06)	586** (94)	7.33** (1.01)	798** (89)	8.00** (1.45)	849** (132)
Ln[maturity, in days]	-1.34** (0.31)	-134** (27)	-0.80* (0.36)	-107** (29)	-1.15** (0.39)	-120** (39)
> 1 tranche indicator	1.73* (0.84)	175* (89)	0.75 (0.66)	84 (64)	1.16 (0.87)	157 (84)

Table IX: public firms requiring monitoring

**Table X
Characteristics of Participants, by Borrowing Firm
Credit Reputation**

This table examines the characteristics of 67,553 participants on 12,672 syndicated loan deals. There are 6,788 participants for private firms, 15,515 participants for unrated firms, and 45,250 for transparent firms. Lead arranger-participant relationships are calculated on a subsample of 43,769 participants on 8,711 deals for which there is exactly one lead arranger.

	Private	Unrated	Transparent
General characteristics			
Total assets (\$B)	243*	235*	295
Equity to total assets ratio	0.067*	0.066*	0.062
Unregulated, domestic (Finance company/I-Bank)	0.069	0.045*	0.064
Foreign	0.40*	0.41*	0.51
Borrowing firm-participant variables			
Conditional on being in the United States:			
In same region as borrowing firm	0.44*	0.43*	0.36
In same census division as borrowing firm	0.32*	0.30*	0.23
In same state as borrowing firm	0.14*	0.13*	0.10
Conditional on firm having previous loan:			
Former lead for firm indicator variable	0.10*	0.11*	0.18
Fraction of previous firm loans lead on	0.05	0.05*	0.06
Conditional on firm having previous syndicated loan:			
Former participant for firm indicator variable	0.50*	0.52*	0.65
Fraction of previous firm syndicated loans participant on	0.34	0.34	0.35
Lead arranger-participant variables			
Calculated on sample with only one lead arranger:			
In same region as lead arranger	0.23*	0.23*	0.20
In same census division as lead arranger	0.16	0.16*	0.14
In same state as lead arranger	0.09	0.09*	0.10
On a deal with lead arranger in previous quarter	0.66*	0.67*	0.77
On a deal with lead arranger in previous year	0.85*	0.85*	0.90
Fraction of syndicated loans with L.A. in previous quarter	0.12*	0.13*	0.15
Fraction of syndicated loans with L.A. in previous year	0.12*	0.13*	0.14

*Significantly different from transparent firms at the 5% level (errors clustered at the firm level)

Table X: participant characteristics

7.6 Tables XI and XII: participant choice probits

Read:

- Table XI estimates the probability that a potential participant is chosen for a deal.
- Prior lead relationships and prior participant relationships strongly increase the probability of being selected, especially for private and unrated borrowers.
- Same-region relationships also increase selection probabilities.
- Table XII repeats the analysis in deals with one lead arranger and finds similar patterns.

Economic message: Syndicates solve information problems not only through lead retention but also through careful participant selection. Relationship lenders and local lenders are especially valuable when borrowers are opaque.

Table XI
Participant Choice Probit Estimation

This table presents coefficient estimates for a probit specification estimating how bank characteristics affect the probability of being chosen as a participant. All coefficients represent the effect on probability when the indicator goes from zero to one, and coefficients and standard errors are multiplied by 100. The choice set includes all banks with at least 0.5% market share in the year of the loan. Estimations include deal level controls described in Table IV and year and industry dummies, and the constant is allowed to vary by group. In addition, the natural log of the total assets and the leverage ratio of the potential participant are included in all specifications and are allowed to vary by group for columns (2) and (4). Columns (2) and (4) have transparent firms as the omitted group. Standard errors are allowed to be correlated for all potential participants for all of a given firm's loans in the sample.

	Without Relationships		With Relationships	
	(1)	(2)	(3)	(4)
<i>Dependent variable mean</i>	8.56	8.56	9.75	9.75
Former lead for borrowing firm indicator			6.40*	6.18*
			(0.69)	(0.67)
Private			3.10*	3.10*
			(1.32)	(1.32)
Unrated			-0.31	-0.31
			(0.60)	(0.60)
Former participant for borrowing firm indicator			27.30*	25.51*
			(0.69)	(0.71)
Private			2.54*	2.54*
			(1.13)	(1.13)
Unrated			2.10*	2.10*
			(0.53)	(0.53)
Same region as borrowing firm indicator	6.68*	5.48*	3.96*	3.23*
	(0.22)	(0.28)	(0.19)	(0.22)
Private		1.46*	1.29*	1.29*
		(0.41)	(0.45)	(0.45)
Unrated		1.94*	1.31*	1.31*
		(0.35)	(0.33)	(0.33)
Foreign indicator	-4.67*	-4.91*	-2.98*	-2.96*
	(0.16)	(0.23)	(0.13)	(0.17)

Table XI: participant choice probit

Table XII
Participant Choice Probits, on Subsample of Deals with One Lead Arranger

This table presents coefficient estimates for a probit specification estimating how bank characteristics affect the probability of being chosen as a participant. The sample is limited to those loans for which there is only one lead arranger. Estimations include deal level controls described in Table IV and year and industry dummies, and the constant is allowed to vary by group. In addition, the natural log of the total assets, the leverage ratio, a foreign indicator variable, and an unregulated domestic indicator for the potential participant are included in all specifications and are allowed to vary by group for columns (2) and (4). Columns (2) and (4) have transparent firms as the omitted group. Standard errors are allowed to be correlated for all potential participants for all of a given firm's loans in the sample.

	Without Relationships		With Relationships	
	(1)	(2)	(3)	(4)
<i>Dependent variable mean</i>	7.98	7.98	9.23	9.23
Former lead for borrowing firm indicator			7.11*	6.77*
			(0.67)	(0.67)
Private			3.28*	3.28*
			(1.32)	(1.32)
Unrated			-0.18	-0.18
			(0.61)	(0.61)
Former participant for borrowing firm indicator			27.48*	24.81*
			(0.72)	(0.77)
Private			2.51	2.51
			(1.53)	(1.53)
Unrated			2.82*	2.82*
			(0.61)	(0.61)
Same region as borrowing firm indicator	6.29*	5.36*	3.77*	3.19*
	(0.22)	(0.31)	(0.20)	(0.24)
Private		1.02*	0.93	0.93
		(0.41)	(0.44)	(0.44)
Unrated		1.18*	0.71	0.71
		(0.34)	(0.32)	(0.32)
Same region as lead arranger	1.16*	1.11*	0.72*	0.59*
	(0.15)	(0.21)	(0.14)	(0.17)

Table XII: one-lead subsample

8 Short summary

- Sufi studies 12,672 syndicated loans from 1992 to 2003.
- The paper treats syndicated loans as an empirical laboratory for information asymmetry and delegated monitoring.
- Borrowers are classified as private, unrated, or transparent.
- Opaque borrowers have syndicates with fewer lenders, higher lead retention, and greater concentration.
- The results are robust to controls for firm size, loan size, maturity, purpose, tranches, and year/industry effects.
- Repeat borrowing reduces opacity problems because borrowers become known to the market.
- Lead arranger reputation mitigates but does not eliminate information asymmetry.
- Collateral does not fully remove the need for lead retention.
- Public firms with higher monitoring needs also have more concentrated syndicates.
- Participant banks are selected partly based on previous relationships and geography.
- The central lesson is that syndicate structure is an endogenous response to monitoring incentives and borrower opacity.

9 Conclusion

9.1 Core conclusion

Sufi shows that information asymmetry shapes both the structure and composition of syndicated loans. When borrowers are opaque, the lead arranger keeps a larger share of the loan and forms a smaller, more concentrated syndicate. This pattern is consistent with moral hazard in monitoring: participant lenders require the lead arranger to retain enough exposure to credibly monitor the borrower.

9.2 Economic intuition

In a syndicated loan, participant lenders rely on the lead arranger’s due diligence and monitoring. The more opaque the borrower, the harder it is for participants to verify that the lead has done this work. A larger retained stake solves part of the problem by making the lead arranger internalize more of the loan’s downside risk. Relationships, reputation, and geographic proximity also help because they provide alternative channels of information.

9.3 Incremental contribution takeaway

The paper’s main incremental contribution is to show that loan syndicate structure itself is a contracting mechanism. It extends delegated monitoring theory by demonstrating that when monitoring is delegated from participant lenders to lead arrangers, the allocation of loan shares, syndicate concentration, lead reputation, borrower reputation, and participant identity all adjust in ways predicted by information-asymmetry theory.